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CS 621 DATA ANALYTICS

Sentiment Analysis and Text Analytics

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SENTIMENT ANALYSIS

- Product reviews are available.
- Each customer's review is represented by a set of words.
- Two interested categories: Positive vs Negative
- Reviews like ["amazing", "fun", "great", "new", …].
- Reviews like ["disappointed", "badly", "loser", …].

- Goal: to "predict/classify/recognize" a new review is positive or negative.

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SENTIMENT ANALYSIS (CONTD)

- A simple yet powerful way to predict a sentiment is to use two pre-defined sets of words – positive vs negative.
- Load positive words as a dictionary (not Python data structure)
- Load negative words as a dictionary.
- Compare each review against these two dictionaries.
- Which score is higher? That is our prediction.
- Check our sample illustration.

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POSITIVE VS NEGATIVE WORDS

```
'assuring',
'astonish',
'astonished',
'astonishing',
'astonishingly',
'astonishment',
'astound',
'astounded',
'astounding',
'astoundingly',
'astutely',
'attentive',
'attraction',
'attractive',
'attractively',
```

```
'destroy',
'destroyer',
'destruction',
'destructive',
'desultory',
'deter',
'deteriorate',
'deteriorating',
'deterioration',
'deterrent',
'detest',
'detestable',
'detestably',
'detested',
```

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SENTIMENT CATEGORIES

- Sometimes, only use “Positive” and “Negative” to label the reviews is not enough.
- How about three sentiment labels?
  - Positive,
  - Negative
  - Neutral
- Emotion categories:
  - Angry
  - Happy
  - Sad
  - Trust
  - …

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EMOJI ANALYSIS

- EMOJIS are commonly used in many social network platforms, .e.g,. Instagram, Tiktok, etc.
- 
- 
- Combined with text analysis.

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NAMED ENTITY RECOGNITION(NER)

- To locate named entities mentioned in the text.
- Pre-defined categories:
  - PERSON: People, including fictional.
  - NORP: Nationalities or religious or political groups.
  - FAC: Buildings, airports, highways, bridges, etc.
  - ORG: Companies, agencies, institutions, etc.
  - GPE: Countries, cities, states.
  - LOC: Non-GPE locations, mountain ranges, bodies of water.
  - PRODUCT: Objects, vehicles, foods, etc. (Not services.)
  - EVENT: Named hurricanes, battles, wars, sports events, etc.
  - WORK_OF_ART: Titles of books, songs, etc.

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NAMED ENTITY RECOGNITION(NER)

- Pre-defined categories:
  - LAW: Named documents made into laws.
  - LANGUAGE: Any named language.
  - DATE: Absolute or relative dates or periods.
  - TIME: Times smaller than a day.
  - PERCENT: Percentage, including ”%“.
  - MONEY: Monetary values, including unit.
  - QUANTITY: Measurements, as of weight or distance.
  - ORDINAL: “first”, “second”, etc.
  - CARDINAL: Numerals that do not fall under another type.

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USING NER IN SPACY

- >>>import spacy
- >>> nlp = spacy.load(‘en_core_web_sm”)
- >>> doc = nlp(message)
- >>> for ent in doc.ents:
  print(ent.text, ent.label_)

See class demo for details.

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NAMED ENTITY RECOGNITION(NER)

Input message:

```
['BREAKING: 6,000 passengers are trapped on a cruise ship after two people on board are tested
/2UbejFI:=:https://abc7chicago.com/health/thousands-stuck-on-cruise-ship-amid-coronavirus-conce
 'BREAKING NEWS: The first case of coronavirus has been confirmed in Orange County, health offi
 'BREAKING: The New Jersey Department of Health has sent out a statement saying that Hackensack
ocal health officials are working together to gather additional information on a woman who is c
ptoms consistent with coronavirus.',
 'CORONAVIRUS LATEST: The outbreak which began in Wuhan, China has claimed the lives of 26 peop
ople.',
 'The World Health Organization has declared the outbreak sparked by a new virus in China that
dozen countries as a global emergency after the number of cases spiked tenfold in a week.']
```

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NAMED ENTITY RECOGNITION(NER)

Output result:

```
6,000 CARDINAL
two CARDINAL
first ORDINAL
Orange County GPE
The New Jersey Department of Health ORG
Hackensack University Medical Center ORG
CORONAVIRUS LATEST ORG
Wuhan GPE
China GPE
26 CARDINAL
more than 800 CARDINAL
The World Health Organization ORG
China GPE
more than a dozen CARDINAL
a week DATE
```



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WE FLY